Assignment 11 Solution

Q.1. Who is credited with developing the Mamdani fuzzy model?

(a) Lotfi A. Zadeh

(b) E. H. Mamdani

(c) John McCarthy

(d) Warren McCulloch

(e) Walter Pitts

Solution: (b)

Q.2. Which components constitute the fundamental elements of a Mamdani fuzzy model?

(a) Evaluating rules

(b) Fuzzifying input variables

(c) Defuzzifying

(d) Aggregating outputs of fuzzy rules

(e) All of the above

Solution: (e)

Q.3. What was the initial application of the Mamdani fuzzy model?

(a) Controlling a steam engine and boiler combination

(b) Regulating a vehicle engine

(c) Creating a golf diagnostic system

(d) Implementing condition-based monitoring for machines

Solution: (a)

Q.4. In the context of a Mamdani fuzzy model described as:

Rule: **IF** x is A **THEN** y is B

Fact (Input): x is A’

Conclusion: y is B’

Which type of Mamdani fuzzy model is this?

(a) Multiple rules with single antecedent

(b) Single rule with multiple antecedents

(c) Single rule with single antecedent

(d) Multiple rules with multiple antecedents

Solution: (c)

Q.5. Let us consider a Mamdani fuzzy model with single rule and multiple antecedents is defined as,

Rule: **IF** x is A and y is B **THEN** z is C

Fact (Input): x is A' and y is B'

Conclusion: z is C'

If the points of intersection for fuzzy inputs A' and B' with A and B, respectively are represented by weights w1=0.70 and w2=0.50, what will be the value of weight obtained using max-min composition for the truncation of C to get the output C'?

(a)0.70

(b)0.50

(c)0.35

(d)None of the above

Solution: (b)

For max-min composition of Mamdani fuzzy model with single rule and multiple antecedents, the weight for the truncation of C to get the output C' is obtained as,

$$w = \min(w1,w2) = \min(0.70,0.50) = 0.50$$

Q.6. For the Mamdani fuzzy model and weights value defined in Q.5., what will be the value of weight obtained using max-product composition for the truncation of C to get the output C'?

(a)0.70

(b)0.50

(c)0.35

(d)None of the above

Solution: (c)

For max-product composition of Mamdani fuzzy model with single rule and multiple antecedents, the weight for the truncation of C to get the output C' is obtained as,

$$w = w1*w2 = 0.70*0.50 = 0.35$$

Q.7. If the aggregated output of a Mamdani fuzzy model is a trapezoidal shaped output defined using parameters [2,5,7,8] with maximum membership function value 1.0, what will be the defuzzified crisp output value using mean of maximum?

(a)6

(b)7

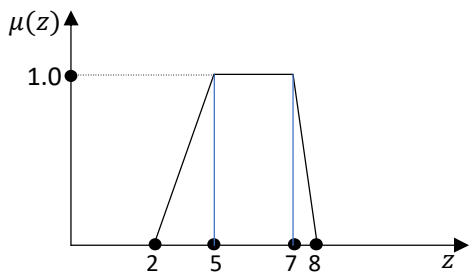
(c)5

(d)2

(e)8

Solution: (a)

Given aggregated output can be plotted as:



The defuzzified crisp output value using mean of maximum will be:

$$z^* = (5+7)/2 = 12/2 = 6$$

Q.8. For the problem statement given in Q.7., what will be defuzzified crisp output value using smallest of maximum?

(a)6

(b)7

(c)5

(d)2

(e)8

Solution: (c)

Q.9. For the problem statement given in Q.7., what will be defuzzified crisp output value using largest of maximum?

(a)6

(b)7

(c)5

(d)2

(e)8

Solution: (b)

Q.10. Which statements accurately describe the defuzzification strategy using the mean of maximum?

(a) The defuzzified value z* is the average of the values of z for which the membership function has its maximum value.

(b) The defuzzified value z* is the average of the values of z for which the membership function has its minimum value.

(c) The defuzzified value z* is the maximum of the values of z.

(d) The defuzzified value $z*$is the minimum of the values of z.

Solution: (a)